COURSEDUP

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Dynamics

Midterm Exam & Review of First Half

INSTRUCTOR	UNIVERSITY	CODE
Leonardo Gonzalez	**Virtual Earth University**	**MCE202**
FACULTY	**DEPARTMENT**	**WEEK**
Science and Engineering	**Mechanical Engineering**	**8**
DISCIPLINE	**TYPE**	**LEVEL**
General	**Midterm Exam**	**Undergraduate**

THEORETICAL FRAMEWORK

- Work-Energy Principle
- Rigid Body Kinematics
- Instantaneous Center of Zero Velocity
- Impulse-Momentum Principle
- Newton's Second Law
- Particle Kinetics
- Equations of Motion
- Acceleration Analysis

Midterm Exam: Principles of Rigid Body Dynamics and Particle Kinetics

Learning Objectives Assessed:

- Perform velocity and acceleration analysis on rigid bodies using the Instantaneous Center of Zero Velocity method.
- Apply Newton's Second Law to solve for forces and motions in complex particle systems.
- Utilize the Work-Energy Principle to determine state changes in both particles and rigid bodies.
- Analyze impulsive forces and momentum changes during particle impact scenarios.
- Derive and solve Equations of Motion for rigid bodies undergoing translation and rotation.

Duration: 90 minutes **Total:** 100 points **Type:** Midterm Exam

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EXAM SCOPE & TOPIC WEIGHTING

This examination is cumulative, encompassing all core concepts introduced from Week 2 through Week 7 of the course. While fundamental principles from early weeks provide the necessary foundation, the assessment is strategically weighted toward the most recent content. Specifically, Weeks 6 and 7 regarding rigid body kinetics represent nearly half of the total marks, reflecting their critical role in engineering analysis. Students should expect a transition from kinematic foundations to more complex kinetic applications as the exam progresses. The design ensures that mastery of early concepts is tested through their application in advanced problem-solving scenarios.

Coverage	Topic(s)	Weight	What to Focus On
Week 7	Kinetics of Rigid Bodies in 2D (Work-Energy Principle for Rigid Bodies)	**24%**	Focus on applying the Work-Energy Principle specifically to rigid bodies involving both translational and rotational kinetic energy.
Week 6	Kinetics of Rigid Bodies in 2D (Equations of Motion for Translation and Rotation)	**20%**	Prioritize the derivation of Equations of Motion for rigid bodies in both translation and rotation.
Week 5	Kinetics of Particles (Impulse-Momentum Principle, Impact)	**17%**	Master the Impulse-Momentum Principle and its application to conservation of momentum during impacts.
Week 4	Kinetics of Particles (Newton's Second Law, Work-Energy Principle)	**15%**	Focus on the systematic application of the Work-Energy Principle to solve particle motion problems.
Week 3	Kinematics of Rigid Bodies in 2D (Acceleration Analysis, Instantaneous Center of Zero Velocity)	**13%**	Practice locating the Instantaneous Center of Zero Velocity to simplify complex acceleration analysis.
Week 2	Kinematics of Rigid Bodies in 2D (Absolute and Relative Motion, Velocity Analysis)	**11%**	Review the vector decomposition methods for absolute and relative velocity in planar mechanisms.

EXAM FORMAT & RULES

The exam is structured into three distinct parts designed to test different levels of cognitive engagement. Part A consists of 10 multiple-choice questions worth 30 points, focusing on conceptual clarity and fundamental definitions. Part B contains 4 short-answer questions worth 40 points, requiring precise calculation and procedural accuracy. Part C features 2 complex scenario analysis questions worth 30 points, which demand the synthesis of multiple concepts. The distribution of marks across these sections is calibrated to mirror the topic weighting, with higher-order scenario questions disproportionately drawing from the critical kinetic principles covered in the latter half of the term.

Section	Question Type	#	Points	Bloom Focus
Part A	Multiple Choice	10	30	Remember/Understand/Apply
Part B	Short Answer	4	40	Apply/Analyze
Part C	Scenario Analysis	2	30	Analyze/Evaluate

Part A: Multiple-choice items covering vocabulary and core concepts across all weeks.

Part B: Short-answer prompts requiring application of concepts.

Part C: Scenario questions integrating evidence across topics.

EXAM RULES

- Ensure all answers are provided in the designated spaces with clear, step-by-step derivations.

- Non-programmable, scientific calculators are permitted for all sections.
- Maintain a clear distinction between vector components in your final solutions.
- Use standard SI units for all numerical results unless otherwise specified.
- Manage your time effectively by allocating minutes proportionally to the point value of each question.
- Ensure all diagrams and free-body diagrams are clearly labeled and aligned with your calculations.

EXAM QUESTION SET

This set matches the structure in the Exam Format table above. Week and difficulty tags show how the topic weighting maps to individual questions.

Part A: Multiple Choice

Choose the single best answer.

1. A rigid body rotates about a fixed axis. If the work done by all external forces is W_{1-2}, which of the following expressions correctly represents the Work-Energy Principle for this body? [Week 7 | Easy]

A. $W_{1-2} = \frac{1}{2} I_G \omega_2^2 - \frac{1}{2} I_G \omega_1^2$

B. $W_{1-2} = \frac{1}{2} I_O \omega_2^2 - \frac{1}{2} I_O \omega_1^2$

C. $W_{1-2} = \frac{1}{2} m v_G^2 - \frac{1}{2} m v_G^2$

D. $W_{1-2} = I_O \alpha$

2. A thin uniform disk of mass m and radius r rolls without slipping down an incline of angle θ. Using the Work-Energy Principle, what is the kinetic energy of the disk after it has rolled a distance s from rest? [Week 7 | Hard]

A. $mgs \sin\theta$

B. $\frac{1}{2} m v^2$

C. $\frac{3}{4} m v^2$

D. $\frac{1}{4} m v^2$

3. For a rigid body in general plane motion, the equations of motion are given by $\sum F_x = m a_{Gx}$, $\sum F_y = m a_{Gy}$, and $\sum M_G = I_G \alpha$. If the body is rotating about a fixed point P, which alternative equation is valid? [Week 6 | Medium]

A. $\sum M_P = I_P \alpha$

B. $\sum M_P = I_G \alpha$

C. $\sum M_P = I_P \alpha + m a_{Gy} d_x$

D. $\sum M_P = I_G \alpha + m a_{Gx} d_y$

4. A slender rod of mass m and length L is released from a horizontal position. What is the initial angular acceleration α as it rotates about a frictionless pin at one end? [Week 6 | Medium]

A. $3g/2L$

B. g/L

C. $6g/L$

D. $g/2L$

5. In a perfectly plastic central impact between two particles, which quantity is conserved? [Week 5 | Medium]

A. Total kinetic energy

B. Total momentum

C. Coefficient of restitution

D. Total energy

6. A ball of mass m strikes a wall with velocity v at an angle θ to the normal and rebounds with velocity v' at angle ϕ. If the coefficient of restitution along the normal is e, what is the normal component of the rebound velocity? [Week 5 | Hard]

A. $ev\cos\theta$

B. $ev\sin\theta$

C. $v\cos\theta$

D. $e^2v\cos\theta$

7. Which of the following is a non-conservative force? [Week 4 | Easy]

A. Gravity

B. Spring force

C. Friction

D. Gravitational potential energy

8. A particle of mass m moves under the influence of a central force. If the particle's speed changes from v_1 to v_2, the work done by the force is: [Week 4 | Medium]

A. $\frac{1}{2}m(v_2^2 - v_1^2)$

B. $\frac{1}{2}m(v_2^2 + v_1^2)$

C. $m(v_2 - v_1)$

D. $mv_2^2 - mv_1^2$

9. For a rigid body undergoing plane motion, the acceleration of a point A relative to a point B is given by $a_A = a_B + \alpha \times r_{A/B} - \omega^2 r_{A/B}$. What does the term $\omega^2 r_{A/B}$ represent? [Week 3 | Hard]

A. Tangential acceleration component

B. Normal acceleration component

C. Coriolis acceleration

D. Total linear acceleration

10. The instantaneous center of zero velocity for a rigid body in plane motion is a point where the velocity is: [Week 2 | Easy]

A. Maximum

B. Infinite

C. Zero

D. Constant

Part B: Short Answer

Answer in the space provided. Show your reasoning clearly.

1. **(10 pts)** A uniform slender rod of mass m and length L is released from rest in a horizontal position. It is attached to a frictionless pivot at one end. Using the Work-Energy Principle, derive the angular velocity ω of the rod when it has rotated through an angle $\theta = 90^\circ$. Compare the work done by gravity to the change in rotational kinetic energy of the rod. [Week 7 | Hard]

2. **(10 pts)** A disk of mass m and radius r rolls without slipping down an incline of angle α. Draw the free-body diagram and the kinetic diagram for the disk. Derive the acceleration of the center of mass a_G using the equations of motion $\sum F = ma_G$ and $\sum M_G = I_G \alpha_{rot}$, where α_{rot} is the angular acceleration. Explain why the friction force is necessary for rolling motion. [Week 6 | Hard]

3. **(10 pts)** A particle of mass m_1 moving with velocity v_1 strikes a stationary particle of mass m_2 in a perfectly plastic impact. After the collision, the particles move together at velocity v_f. Derive an expression for the energy loss ΔE during the impact in terms of m_1, m_2, and v_1. Discuss the conservation laws applicable during this process. [Week 5 | Hard]

4. **(10 pts)** A block of mass m slides down a rough incline with kinetic coefficient of friction μ_k. The block starts from rest at the top of the incline of length L and angle θ. Use the Work-Energy Principle to determine the velocity of the block at the bottom of the incline. If a spring of constant k is placed at the bottom to stop the block, determine the maximum compression x of the spring using the same principle. [Week 4 | Hard]

Part C: Scenario Analysis

Answer in the space provided. Show your reasoning clearly.

1. **(15 pts)** A uniform slender rod of mass m and length L is released from rest in a horizontal position. The rod is pinned at one end O and rotates in a vertical plane. A spring with stiffness k is attached to the other end of the rod, initially unstretched. Determine the angular velocity ω of the rod when it has rotated through an angle $\theta = 90^\circ$ using the Work-Energy Principle. Account for the gravitational potential energy and the elastic potential energy stored in the spring. [Week 7 | Hard]

2. **(15 pts)** A solid cylinder of mass m and radius r rolls without slipping down an incline of angle α. Using the equations of motion for rotation and translation, derive an expression for the acceleration of the center of mass a_G and the required coefficient of static friction μ_s to ensure no slipping occurs. Assume the cylinder has a moment of inertia $I_G = \frac{1}{2}mr^2$. [Week 6 | Hard]

HOW TO PREPARE — STUDY RECOMMENDATIONS

- Allocate 50% of your study time to the rigid body kinetics covered in Weeks 6 and 7 to align with the exam weighting.
- Create a summary sheet of all governing equations for Work-Energy and Impulse-Momentum to identify patterns across different problem types.
- Practice solving multi-step problems without looking at solutions to build the stamina required for the 10/10 difficulty level.
- Review your past assignments to identify common errors in sign conventions during acceleration analysis.
- Use metacognitive reflection to explain why you chose a specific principle (e.g., Work-Energy vs. Newton's Law) for a given scenario.

INSTRUCTOR COPY — ANSWER KEY & MODEL ANSWERS

Part A — MCQ Answer Key

Q	Answer	Q	Answer
1	**B**	6	**A**
2	**A**	7	**C**
3	**A**	8	**A**
4	**A**	9	**B**
5	**B**	10	**C**

Solution Explanations:

- **Q1 (B):** For rotation about a fixed axis O, the kinetic energy is $\frac{1}{2}I_O\omega^2$. Option A is incorrect because it uses the centroidal moment of inertia instead of the axis of rotation, which is only valid if O is G.
- **Q2 (A):** The work done by gravity is $mgs\sin\theta$. Since the disk starts from rest, this work equals the final kinetic energy. Distractor C represents the total kinetic energy formula $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$, but the question asks for the value of the energy itself, which is simply the work done.
- **Q3 (A):** When a body rotates about a fixed point P, the moment equation about that point is $\sum M_P = I_P\alpha$. Option B is a common error as it neglects the parallel axis theorem term md^2 implicitly contained in I_P.
- **Q4 (A):** Using $\sum M_P = I_P\alpha$, we have $mg(L/2) = (\frac{1}{3}mL^2)\alpha$. Solving for α gives $3g/2L$. Distractor B is incorrect because it uses the wrong moment of inertia for a rod about the end.
- **Q5 (B):** Total momentum is conserved in all impacts. Kinetic energy is only conserved in perfectly elastic impacts ($e = 1$). In plastic impacts, kinetic energy is lost.
- **Q6 (A):** The normal component of the initial velocity is $v_n = v\cos\theta$. By definition of the coefficient of restitution, $v'_n = ev_n = ev\cos\theta$. Distractor B uses the wrong trigonometric function.
- **Q7 (C):** Friction is a non-conservative force because the work done depends on the path taken. Gravity and spring forces are conservative.
- **Q8 (A):** By the Work-Energy Theorem, the work done is equal to the change in kinetic energy $\Delta T = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$. Distractor D misses the factor of 1/2.
- **Q9 (B):** The term $\omega^2 r_{A/B}$ is the normal (centripetal) acceleration directed from A toward B. Distractor A is represented by $\alpha \times r_{A/B}$.
- **Q10 (C):** By definition, the instantaneous center is the point about which the body appears to be rotating at a given instant, meaning its velocity is zero. Distractor D is incorrect as the velocity of other points is not constant.

Part B — Model Answers

Q1 (10 pts): A uniform slender rod of mass m and length L is released from rest in a horizontal position. It is attached to a frictionless pivot at one end. Using the Work-Energy Principle, derive the angular velocity ω of the rod when it has rotated through an angle $\theta = 90°$. Compare the work done by gravity to the change in rotational kinetic energy of the rod.

Model Answer / Key Points: 1. Identify the potential energy change: $\Delta V_g = -mg(L/2)$. 2. Identify the kinetic energy change: $\Delta T = 1/2I_O\omega^2 - 0$. 3. Use mass moment of inertia $I_O = 1/3mL^2$. 4. Apply $T_1 + V_1 = T_2 + V_2$: $0 + 0 = 1/2(1/3mL^2)\omega^2 - mg(L/2)$. 5. Solve for ω: $1/6mL^2\omega^2 = mgL/2 \implies \omega = \sqrt{3g/L}$. The work done by

gravity equals the change in kinetic energy.

Q2 (10 pts): A disk of mass m and radius r rolls without slipping down an incline of angle α. Draw the free-body diagram and the kinetic diagram for the disk. Derive the acceleration of the center of mass a_G using the equations of motion $\sum F = ma_G$ and $\sum M_G = I_G\alpha_{rot}$, where α_{rot} is the angular acceleration. Explain why the friction force is necessary for rolling motion.

Model Answer / Key Points: 1. FBD shows weight mg, normal force N, and friction f. 2. $\sum F_x = mg\sin\alpha - f = ma_G$. 3. $\sum M_G = fr = I_G\alpha_{rot}$. 4. Kinematic constraint: $a_G = r\alpha_{rot}$. 5. Substitute $f = I_G(a_G/r^2)$ into the force equation: $mg\sin\alpha - (I_G a_G/r^2) = ma_G$. 6. With $I_G = 1/2mr^2$, $a_G = (mg\sin\alpha)/(m + 1/2m) = 2/3g\sin\alpha$. Friction provides the torque required to produce angular acceleration.

Q3 (10 pts): A particle of mass m_1 moving with velocity v_1 strikes a stationary particle of mass m_2 in a perfectly plastic impact. After the collision, the particles move together at velocity v_f. Derive an expression for the energy loss ΔE during the impact in terms of m_1, m_2, and v_1. Discuss the conservation laws applicable during this process.

Model Answer / Key Points: 1. Conservation of linear momentum: $m_1v_1 = (m_1 + m_2)v_f \implies v_f = [m_1/(m_1 + m_2)]v_1$. 2. Initial kinetic energy $T_i = 1/2m_1v_1^2$. 3. Final kinetic energy $T_f = 1/2(m_1 + m_2)v_f^2 = 1/2(m_1 + m_2)[m_1/(m_1 + m_2)]^2v_1^2 = 1/2[m_1^2/(m_1 + m_2)]v_1^2$. 4. Energy loss $\Delta E = T_i - T_f = 1/2m_1v_1^2[1 - m_1/(m_1 + m_2)] = 1/2[m_1m_2/(m_1 + m_2)]v_1^2$. Momentum is conserved; energy is not conserved due to plastic deformation.

Q4 (10 pts): A block of mass m slides down a rough incline with kinetic coefficient of friction μ_k. The block starts from rest at the top of the incline of length L and angle θ. Use the Work-Energy Principle to determine the velocity of the block at the bottom of the incline. If a spring of constant k is placed at the bottom to stop the block, determine the maximum compression x of the spring using the same principle.

Model Answer / Key Points: 1. Work done by gravity: $W_g = mgL\sin\theta$. 2. Work done by friction: $W_f = -f_kL = -\mu_k(mg\cos\theta)L$. 3. Work-Energy: $T_1 + \sum U_{1-2} = T_2 \implies 0 + mgL\sin\theta - \mu_kmgL\cos\theta = 1/2mv^2$. 4. Velocity $v = \sqrt{2gL(\sin\theta - \mu_k\cos\theta)}$. 5. For spring compression: The total work done equals the change in potential energy of the spring $1/2kx^2$. $mg(L + x)\sin\theta - \mu_kmg\cos\theta(L + x) = 1/2kx^2$. Solve the resulting quadratic for x.

Part C — Model Answers

Q1 (15 pts): A uniform slender rod of mass m and length L is released from rest in a horizontal position. The rod is pinned at one end O and rotates in a vertical plane. A spring with stiffness k is attached to the other end of the rod, initially unstretched. Determine the angular velocity ω of the rod when it has rotated through an angle $\theta = 90^\circ$ using the Work-Energy Principle. Account for the gravitational potential energy and the elastic potential energy stored in the spring.

Model Answer / Key Points: 1. Define the system state at position 1 (horizontal) and position 2 (vertical). Position 1: $T_1 = 0$, $V_{g1} = 0$, $V_{e1} = 0$. Position 2: $T_2 = \frac{1}{2}I_O\omega^2$, where $I_O = \frac{1}{3}mL^2$. 2. Gravitational potential change: $\Delta V_g = -mg(L/2)$. 3. Spring potential energy: The displacement of the spring end is $s = L\theta = L(\pi/2)$. Thus, $V_{e2} = \frac{1}{2}k(L\pi/2)^2$. 4. Apply $T_1 + V_1 = T_2 + V_2$: $0 = \frac{1}{2}(\frac{1}{3}mL^2)\omega^2 - mg(L/2) + \frac{1}{2}k(L^2\pi^2/4)$. 5. Solve for ω: $\omega = \sqrt{\frac{3g}{L} - \frac{3kL\pi^2}{4m}}$.

Q2 (15 pts): A solid cylinder of mass m and radius r rolls without slipping down an incline of angle α. Using the equations of motion for rotation and translation, derive an expression for the acceleration of the center of mass a_G and the required coefficient of static friction μ_s to ensure no slipping occurs. Assume the cylinder has a moment of inertia $I_G = \frac{1}{2}mr^2$.

Model Answer / Key Points: 1. Free body diagram: Gravity mg (downward), Normal force N (perpendicular to incline), Friction F (up the incline). 2. Equations of motion: $\sum F_x = mg\sin\alpha - F = ma_G$; $\sum F_y = N - mg\cos\alpha = 0$; $\sum M_G = Fr = I_G\alpha_{rot}$. 3. Kinematic constraint: $a_G = r\alpha_{rot}$, so $F = I_G a_G/r^2 = (\frac{1}{2}mr^2)a_G/r^2 = \frac{1}{2}ma_G$. 4. Substitute F into force equation: $mg\sin\alpha - \frac{1}{2}ma_G = ma_G \Rightarrow a_G = \frac{2}{3}g\sin\alpha$. 5. Find friction: $F = \frac{1}{3}mg\sin\alpha$. 6. Condition for no slip: $F \leq \mu_s N \Rightarrow \frac{1}{3}mg\sin\alpha \leq \mu_s mg\cos\alpha \Rightarrow \mu_s \geq \frac{1}{3}\tan\alpha$.